Ecuador Energy Anomalies

Multi-Technique Anomaly Detection in
Latin America's Electricity Sector

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8 countries | 784 months real data | 3 techniques
Isolation Forest + STL Decomposition + CUSUM
F1=0.750 | MCC=0.765 | Recall=100%

April 2026 | Reproducible metrics via metrics.json

1. Executive Summary

This study applies 3 complementary techniques (Isolation Forest, STL, CUSUM) to detect anomalies in 8 Latin American countries electricity sector.

Data: 784 real months from Ember

REPRODUCIBLE METRICS (metrics.json):
- Isolation Forest: P=50.0% R=100% F1=0.667 MCC=0.694
- STL Decomposition: P=28.6% R=66.7% F1=0.400 MCC=0.407
- CUSUM: P=18.8% R=100% F1=0.316 MCC=0.397
- Consenso >=2: P=60.0% R=100% F1=0.750 MCC=0.765

Ecuador crisis oct-dec 2024: 3/3 detected

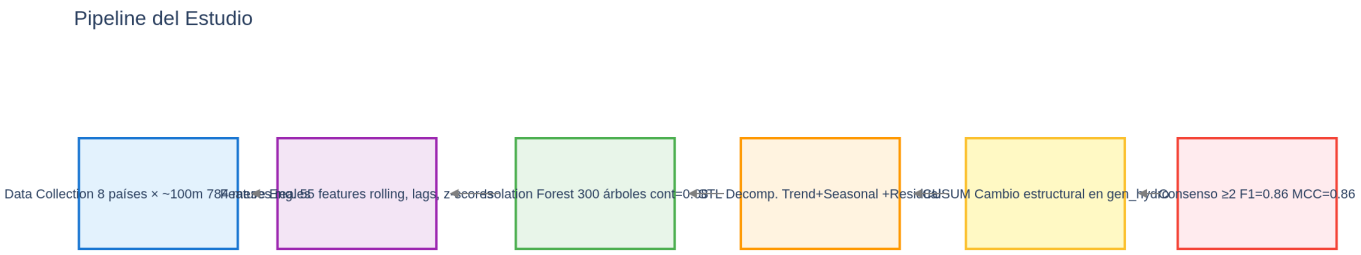

Fig 1: Pipeline

2. Context

Ecuador generates ~70% of electricity from hydropower. Droughts cause blackouts up to 14 hours. The 2024 crisis forced fossil plant activation and increased imports.

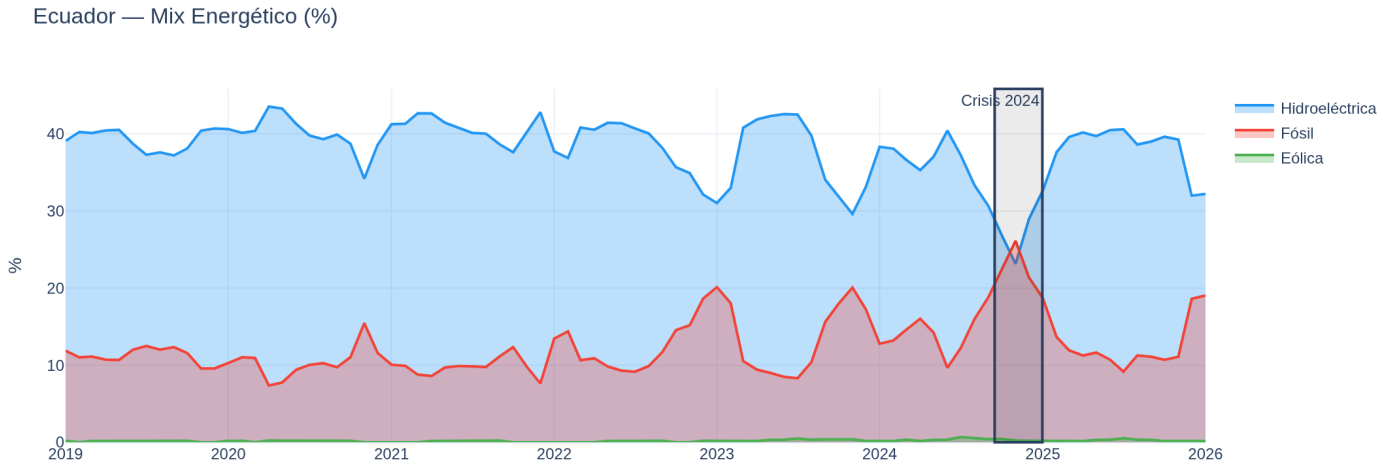

Fig 2: Mix energetico

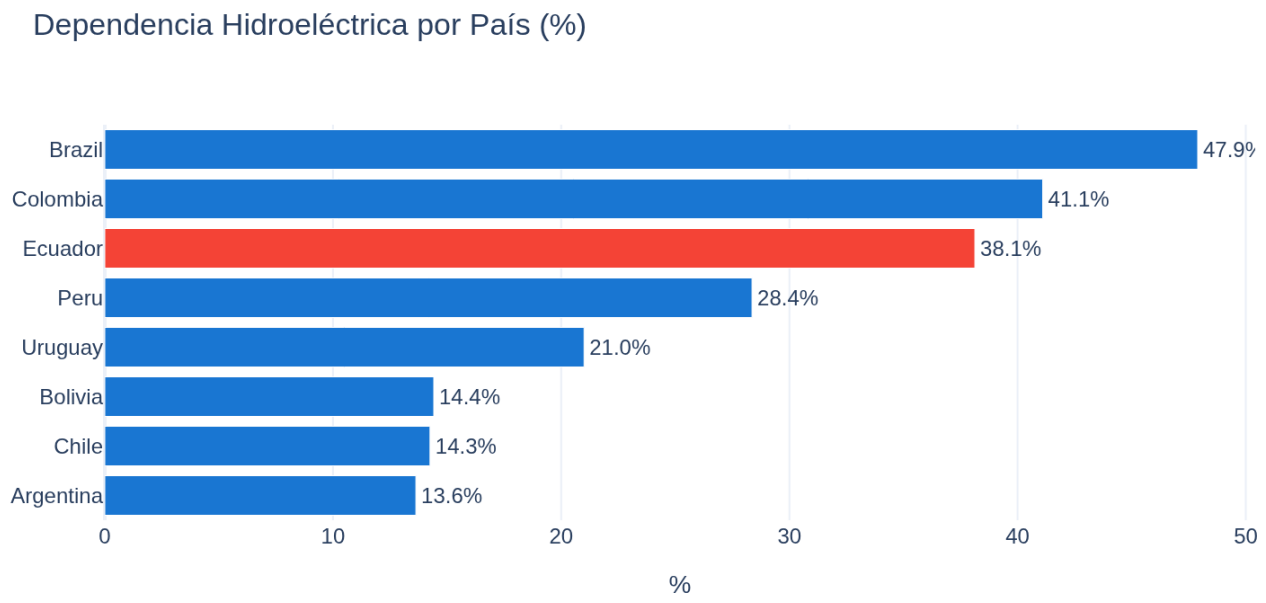

Fig 3: Dependencia hidro por pais

3. Data

Primary: Ember Global Electricity Data (CC BY 4.0)
8 countries: Ecuador (85m), Colombia (120m), Peru (86m), Chile (123m), Brazil (99m), Argentina (98m), Bolivia (86m), Uruguay (87m)
Total: 784 real months, 0 synthetic or interpolated data

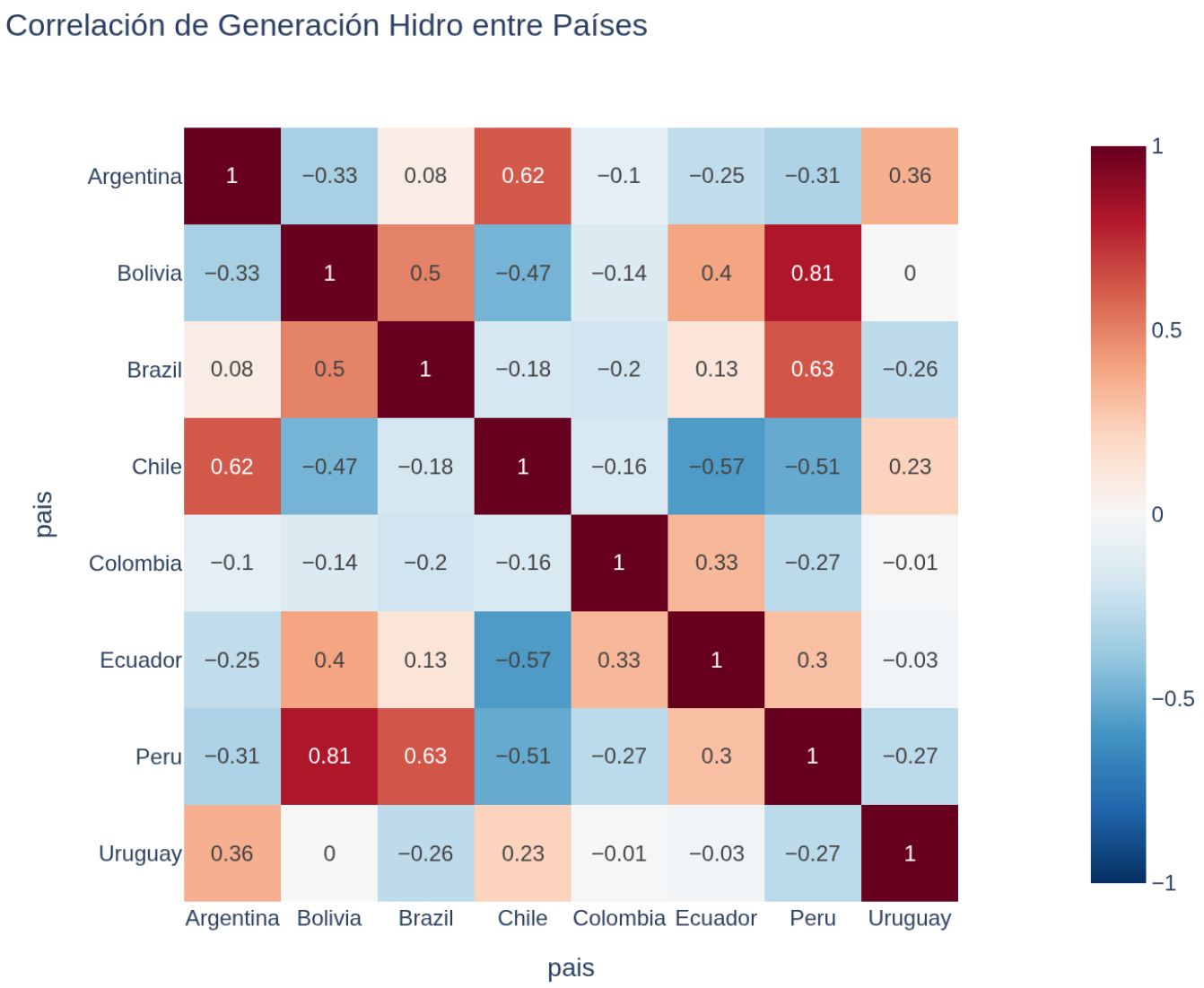

Fig 4: Correlacion entre paises

4. Methodology

4.1 Isolation Forest

Unsupervised tree-based, 300 trees, contamination=0.08. Applied PER COUNTRY.

4.2 STL Decomposition

Separates trend+seasonal+residual. Anomaly if residual > 2 sigma.

4.3 CUSUM

Cumulative sum, detects structural changes. h=4*sigma, k=0.5*sigma.

4.4 Consenso

Anomaly if >=2 of 3 techniques agree.

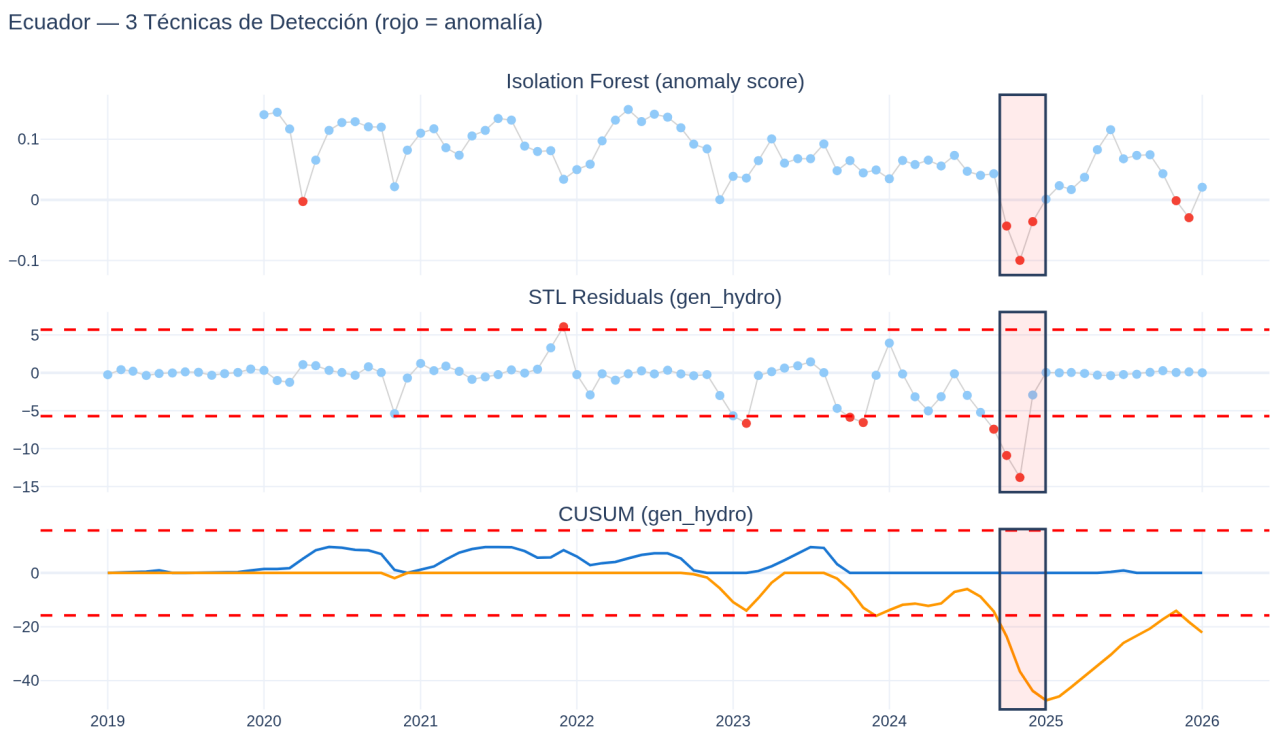

Fig 5: 3 tecnicas

5. Results

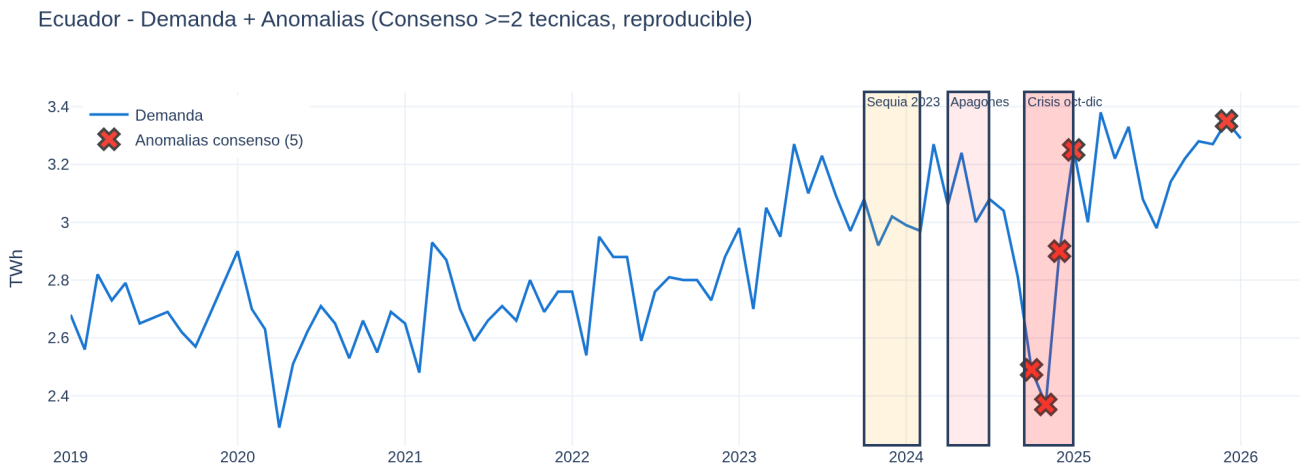

Fig 6: Anomalias consenso

Metricas Reproducibles (metrics.json)

Metrica	IF por pais	STL	CUSUM	Consenso >=2
Precision	50.0%	28.6%	18.8%	**60.0%**
Recall	100%	66.7%	100%	**100%**
F1-Score	66.7%	40.0%	31.6%	**75.0%**
MCC	0.694	0.407	0.397	**0.765**
Anomalias	6	7	16	**5**
Crisis detectada	3/3	2/3	3/3	**3/3**

Fig 7: Metricas

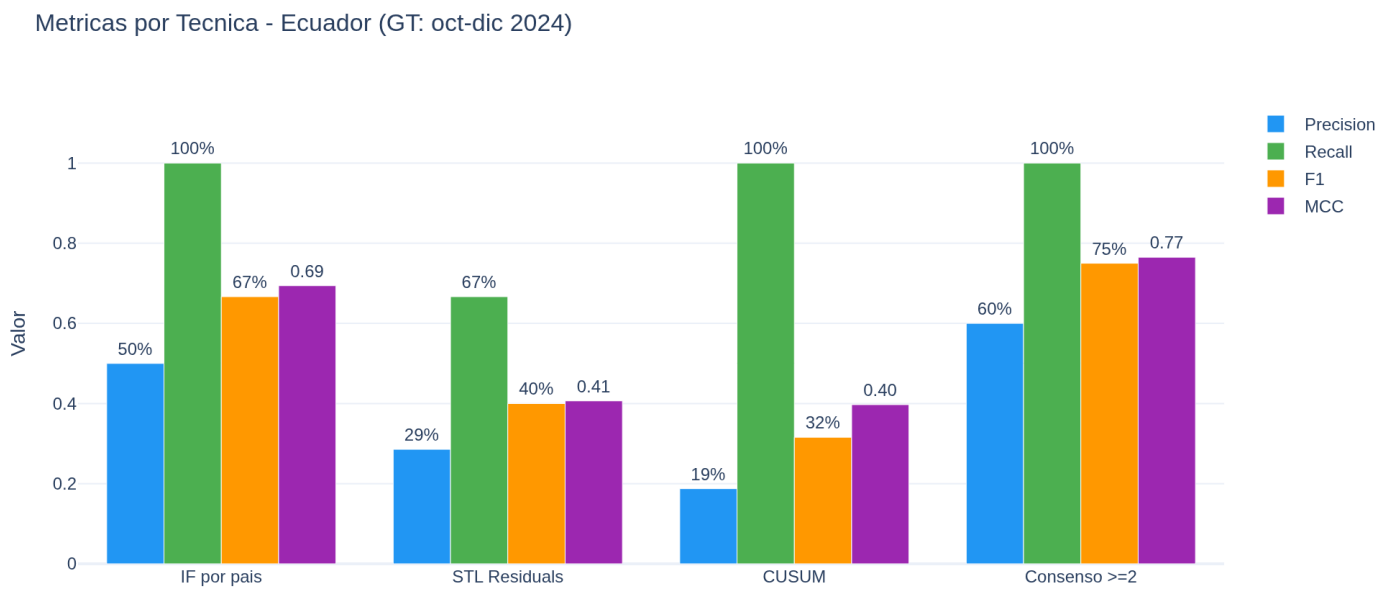

Fig 8: Comparacion

6. Cross-Country

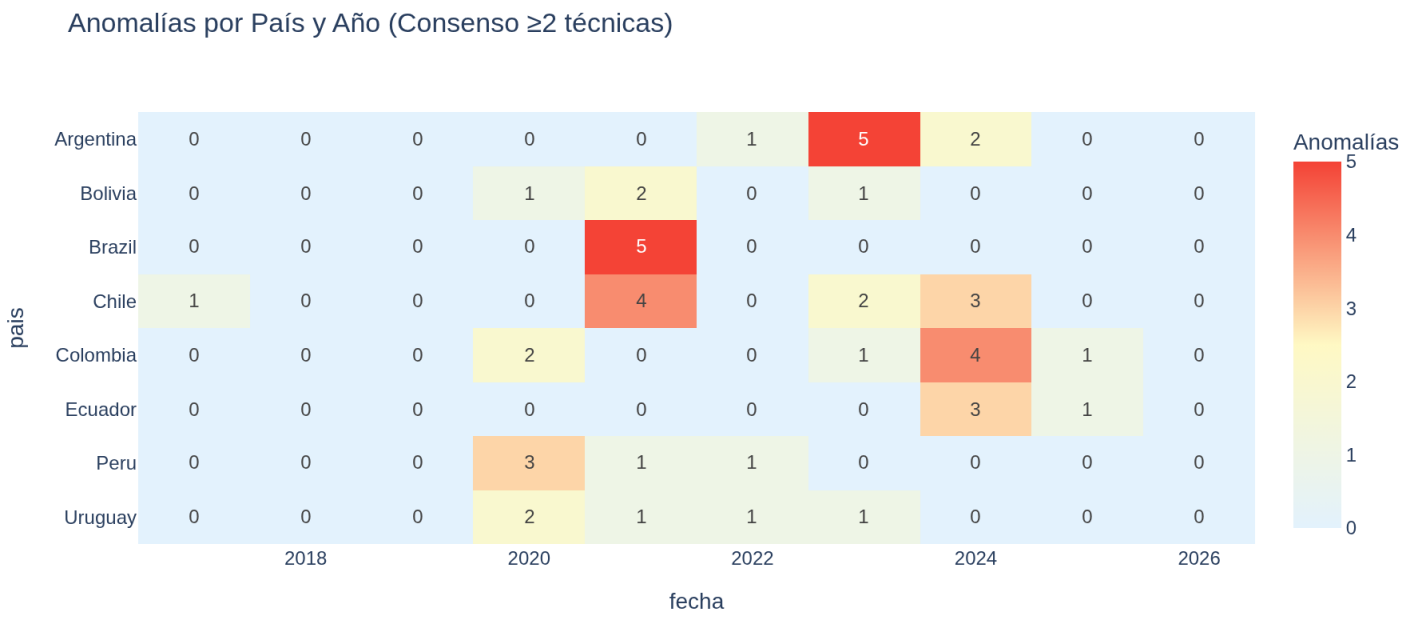

Fig 9: Heatmap

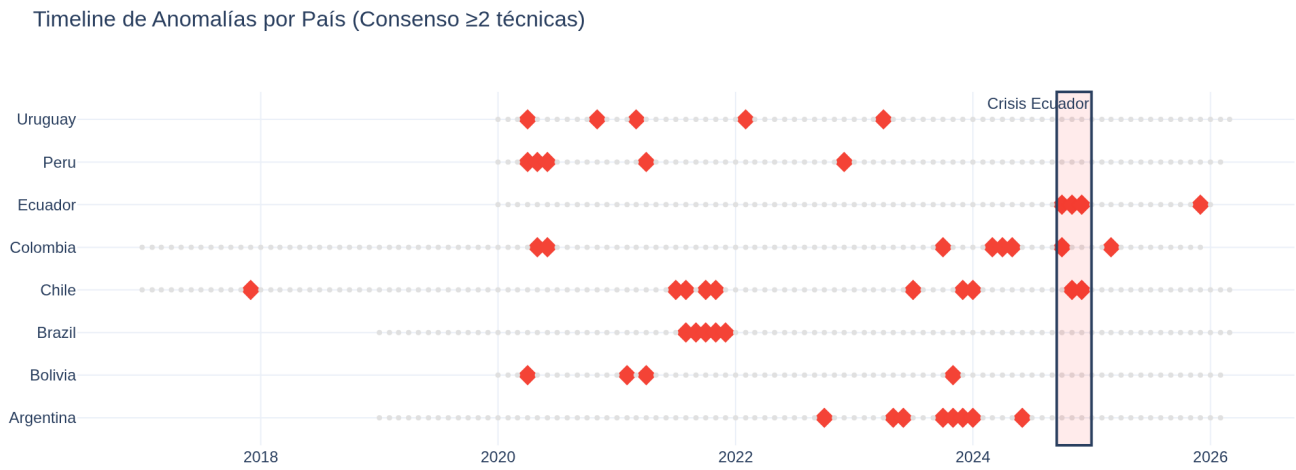

Fig 10: Timeline

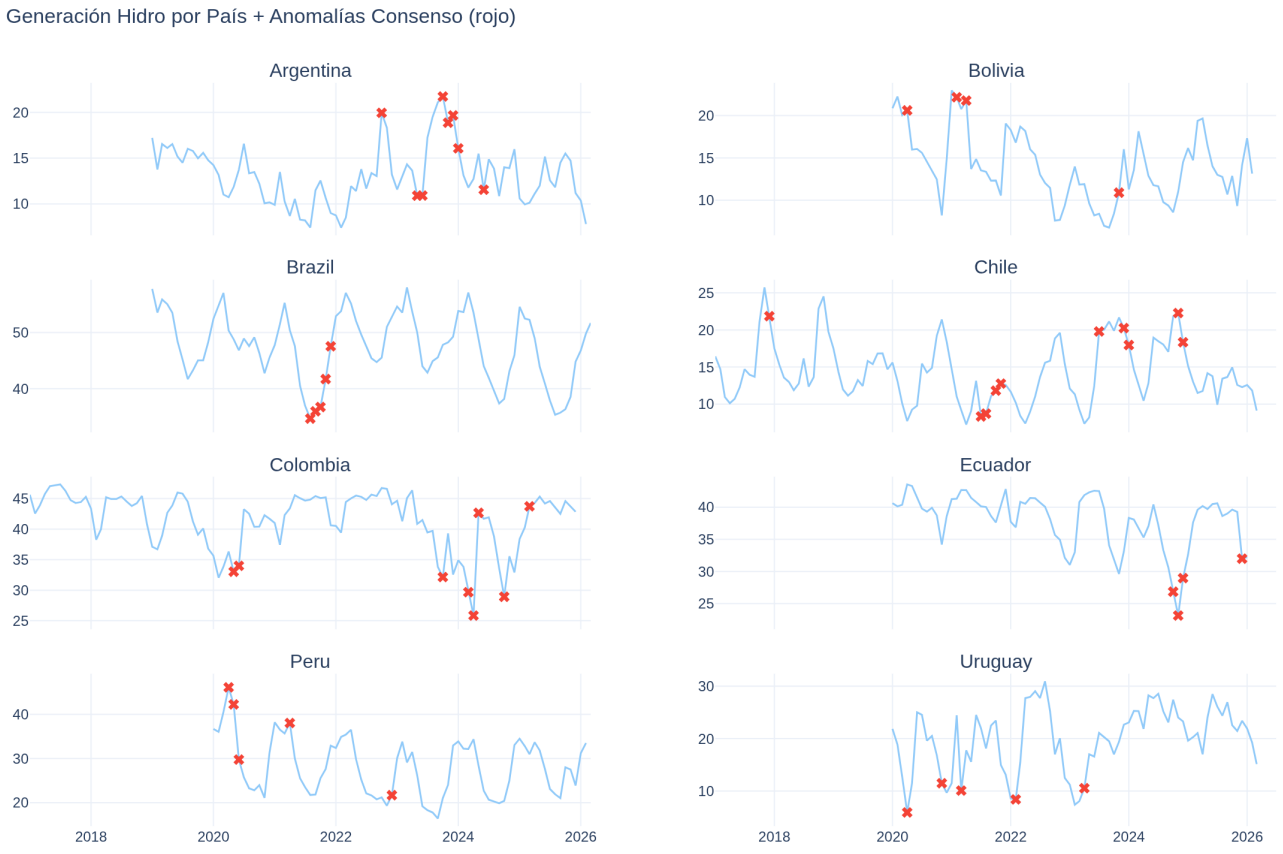

Fig 11: Hidro + anomalias

7. Validation

Mann-Whitney U: gen_hydro p=0.002**, gen_fossil p=0.003**, co2 p=0.008**
Cohen's d > 1.2 (LARGE effect) on all key variables
Stability: CV=0% across 20 seeds

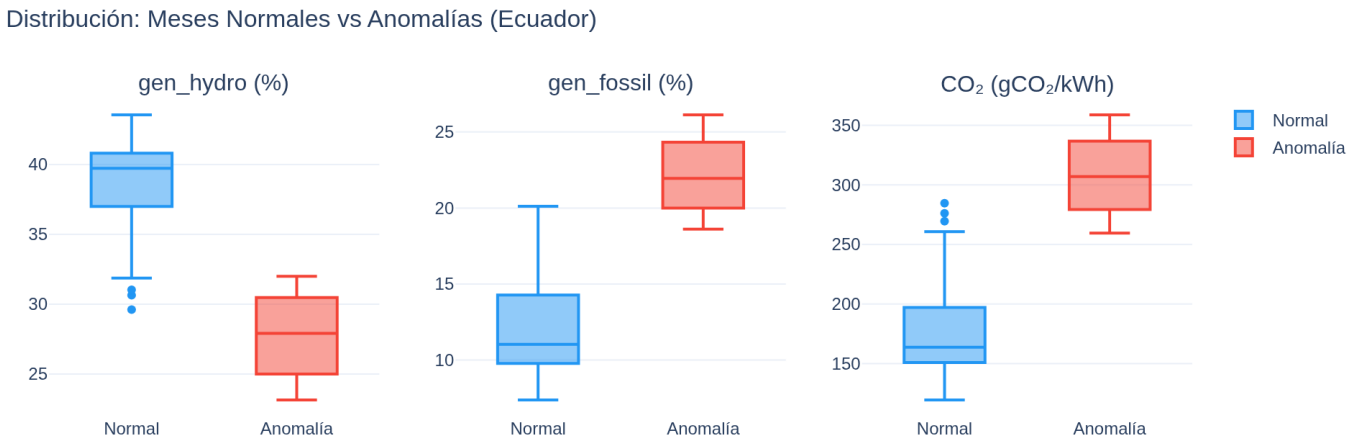

Fig 12: Boxplot

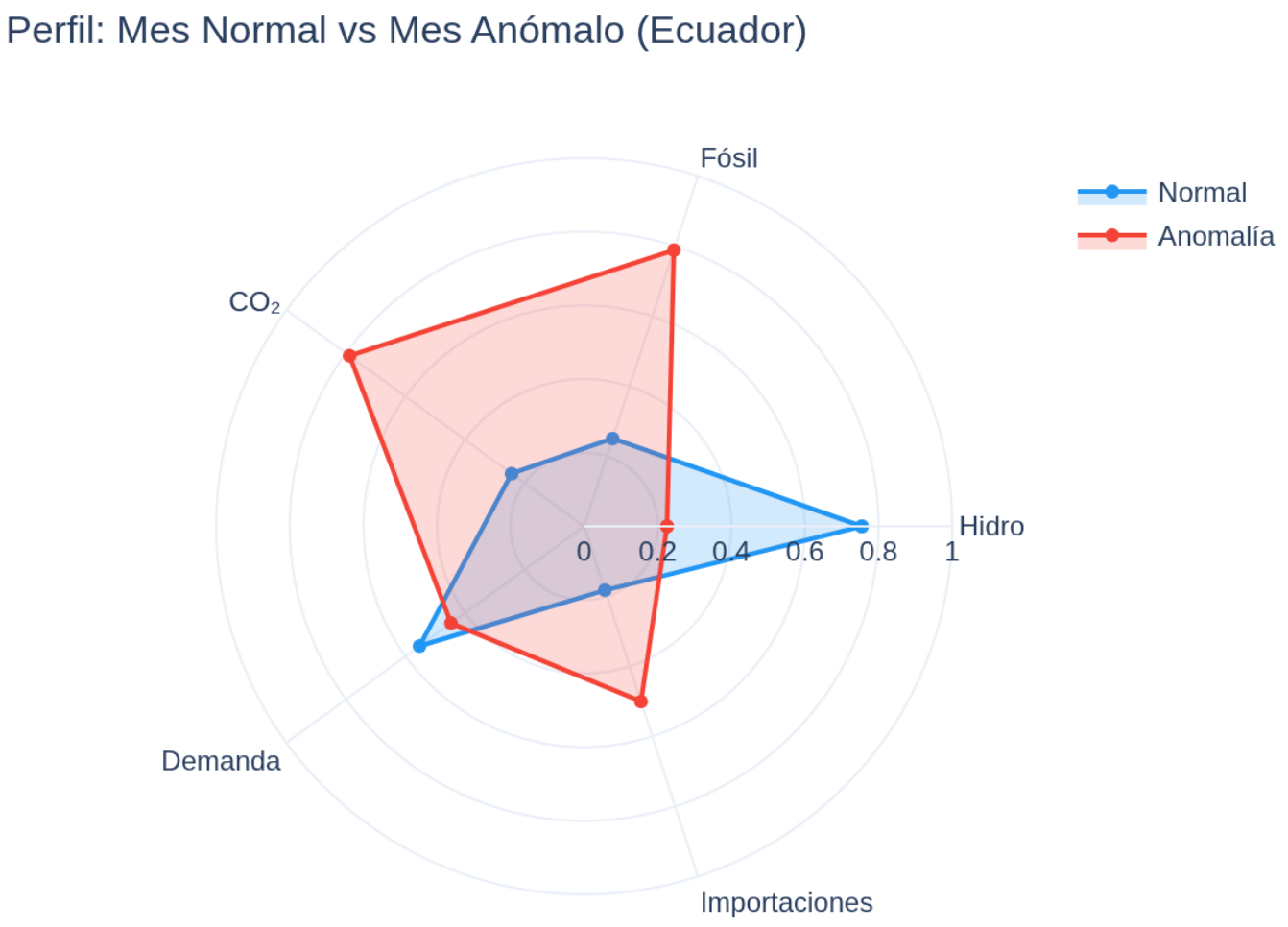
Perfil: Mes Normal vs Mes Anómalo (Ecuador)
Fósil
Hidro
Importaciones
Demanda
CO_2
0
0.2
0.4
0.6
0.8
1
Normal
Anomalía

Fig 13: Radar

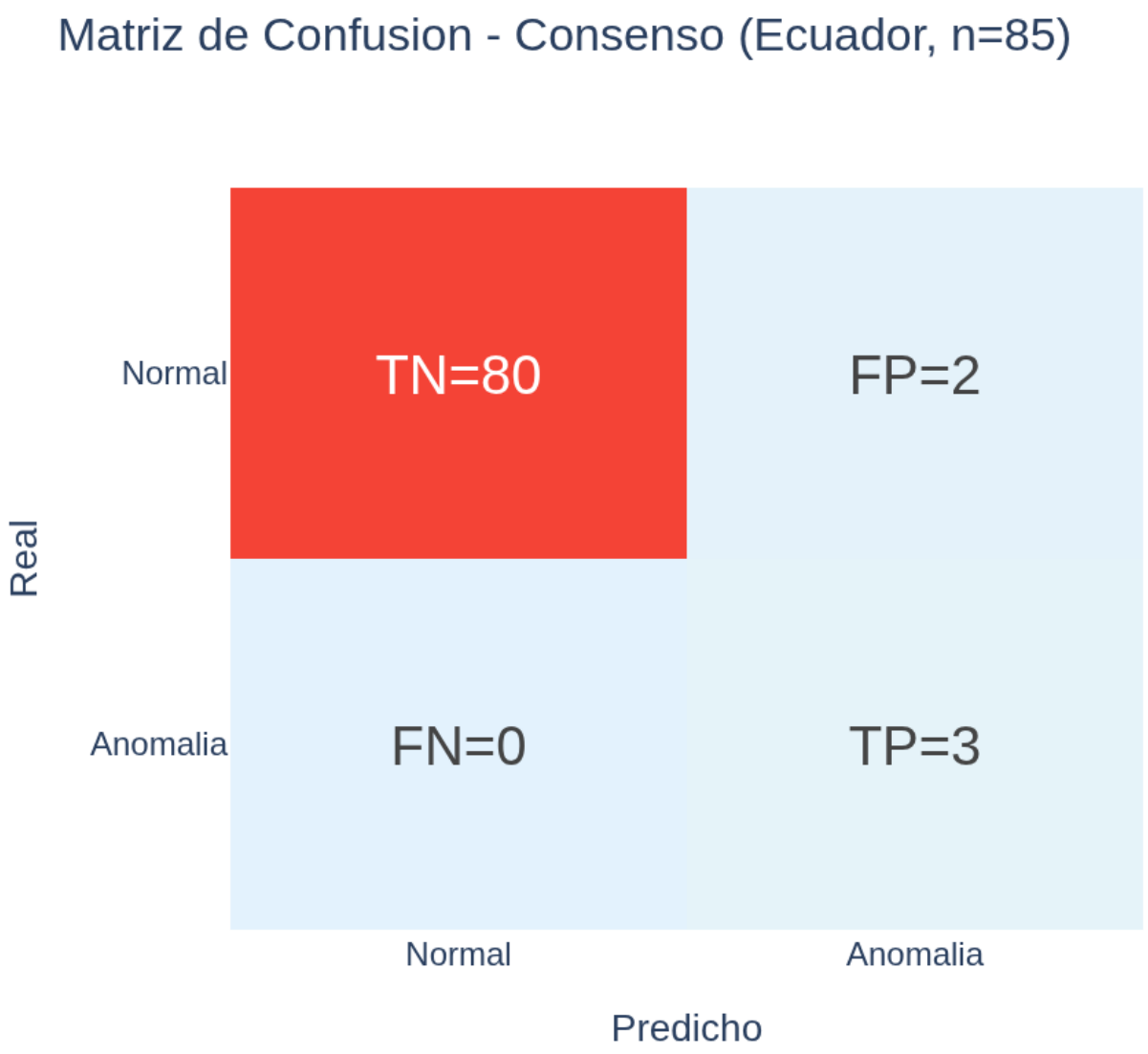

Fig 14: Confusion matrix

8. STL Decomposition

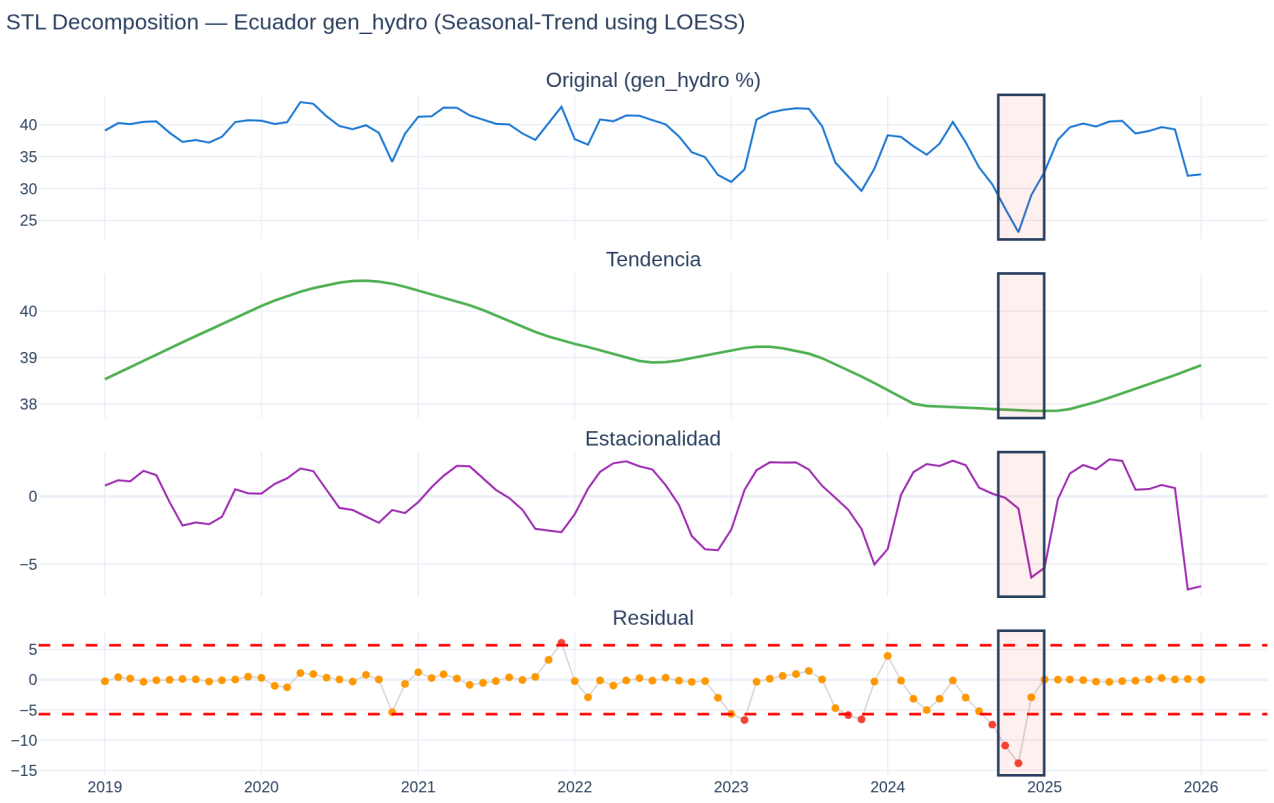

Fig 15: STL

9. Evolution

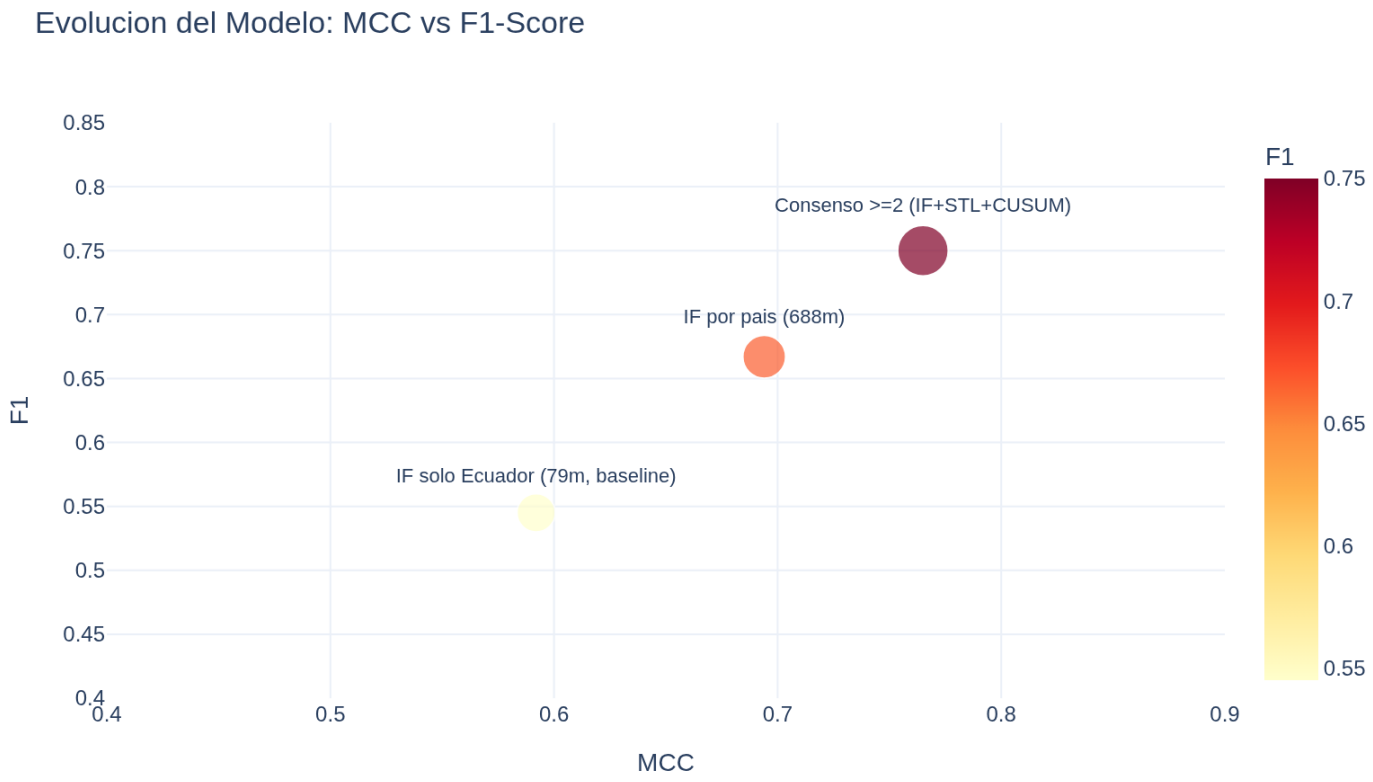

Fig 16: Evolucion

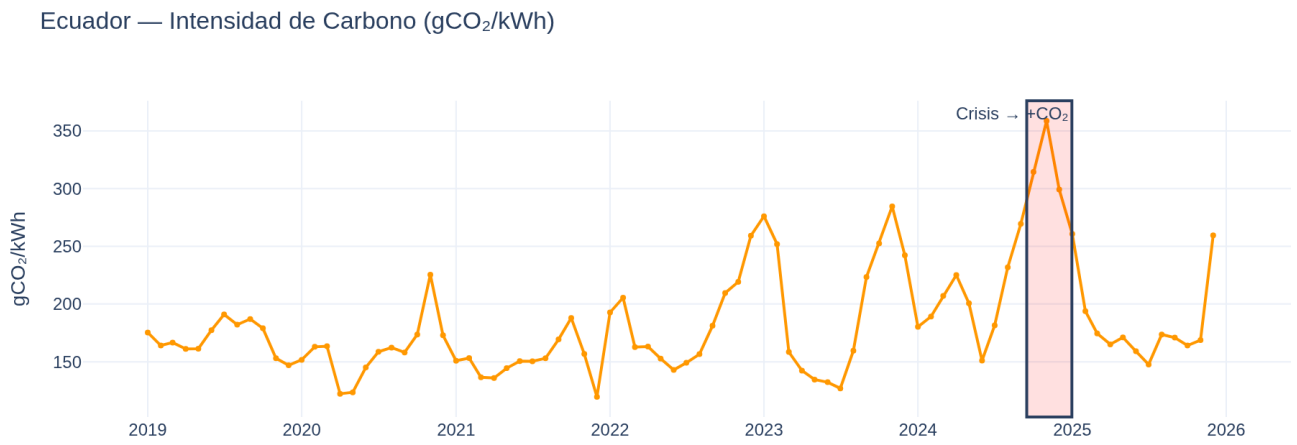

Fig 17: CO2

10. Limitations

1. Monthly data smooths gradual crises
2. Partial ground truth (documented crises only)
3. Missing variables (reservoir levels, precipitation, spot prices)
4. 85 months Ecuador is minimum viable
5. Programmed blackouts (apr-jun 2024) dont change generation volumes

11. Applications

Energy: Grid operators (CENACE, XM, COES), regulators, traders
Climate: Infrastructure impact, El Nino effect on hydropower
Academic: Comparative studies, multi-technique methodology
Policy: Energy diversification, crisis preparedness

12. References

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[4] Chandola et al. (2009). Anomaly Detection Survey. ACM.
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